UNIVERSITY OF
PERPETUAL HELP
SYSTEM DALTA

GradeSense AI:
An Intelligent Automated Grading
and
Feedback System for Modern Education

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Project Context

In modern education, **grading** and **feedback** are still often performed manually by teachers, making the process time-consuming, labor-intensive, and sometimes inconsistent. In many cases, teachers handle large class sizes, which further increases the burden of checking assessments and providing individualized feedback. As a result, feedback is often delayed, limiting students' ability to reflect on their mistakes and improve their performance in a timely manner.

With the advancement of Artificial Intelligence (AI), there is an opportunity to enhance the grading process by making it faster, more consistent, and efficient. AI technologies can analyze patterns, evaluate responses, and generate insights that can support educators in their tasks. GradeSense AI is proposed as a solution to assist educators in automating grading tasks while providing immediate, reliable, and meaningful feedback to students, ultimately improving the overall teaching and learning experience.

PURPOSE AND DESCRIPTION

The purpose of GradeSense AI is to develop an automated grading system capable of evaluating various types of student outputs, such as quizzes, exams, and essays. The system aims to reduce the time teachers spend on repetitive grading tasks while ensuring accuracy and consistency in evaluation.

The system will utilize AI techniques, particularly machine learning and natural language processing (NLP), to:

- Automatically grade objective and subjective assessments with minimal human intervention
- Provide instant, detailed, and constructive feedback to guide student learning
- Suggest improvements and highlight areas of weakness based on student performance
- Adapt to different question formats and difficulty levels over time

It will feature a user-friendly and accessible interface where teachers can upload assessments, manage classes, and review results, while students can easily view their grades, feedback, and progress. The platform will also aim to promote a more interactive and responsive learning environment.

General Objective:

To develop an AI-powered grading and feedback system that improves efficiency, consistency, and quality in student evaluation while supporting both teachers and learners in the academic process.

Specific Objective:

- To design a system that can automatically and accurately grade multiple-choice and essay-type assessments
- To provide real-time and actionable feedback to students immediately after submission
- To store, organize, and track student performance data over time for progress monitoring
- To generate personalized suggestions for improvement based on individual results and trends
- To create an intuitive, responsive, and user-friendly interface for both teachers and students
- To ensure data privacy and security in handling student records and assessment results

Scope & Limitations

Scope:

- Focuses on grading quizzes, exams, and basic written outputs (e.g., short essays and reflections)
- Web-based application accessible to both teachers and students using standard devices
- Includes user authentication (login system) to ensure secure access
- Provides dashboards for grades, analytics, feedback, and performance tracking
- Supports basic data visualization for easier interpretation of student progress

Limitations:

- Essay evaluation may be limited primarily to the English language and may not fully capture nuanced writing styles
- Not designed for complex research papers, highly technical writing, or creative works requiring deep human interpretation
- AI accuracy depends on the quality, diversity, and quantity of training data used
- The system is intended to assist teachers, not replace human judgment in grading and evaluation
- Requires stable internet connection for optimal performance and accessibility

Significance of the Study

GradeSense AI contributes to the improvement of educational assessment by addressing the challenges of manual grading and delayed feedback. Through the integration of artificial intelligence, the system enhances the efficiency, consistency, and accuracy of evaluating student outputs, allowing teachers to focus more on instruction and student engagement.

For educators, the system reduces workload by automating routine grading tasks while maintaining reliable evaluation standards. For students, it provides immediate and constructive feedback, enabling them to identify areas for improvement and support their learning progress more effectively.

At the institutional level, **GradeSense AI** promotes standardized and data-driven assessment practices, improving overall academic management. Additionally, this study serves as a foundation for future advancements in AI-based educational technologies, particularly in automated evaluation and feedback systems.